Advanced Computer-Aided VLSI System Design

Low-Power Design Lab: UPF Flow

Objectives:

In this lab, you will learn:

How to add low-power design constraint through UPF flow.

Copy the reference ADFP lab package

```
➢ cp -rf /share1/tech/ADFP/ ~/local_adfp_lab
```

Environment Setup

Type below command to source the required cell-based CAD tools:

```
➢ cb
```

Modify the project root of the lab in var.tcl. In the Path: ~/local_adfp_lab/Executable_Package/Lab/Work/sub_block/01_rtl_to_netlist/scr/**var.tcl**

```
➢ set ProjRoot "Your Absolute Path for local_adfp_lab"
```

For example:

```
➢ set ProjRoot "/share1/home/acvsd-ta02/local_adfp_lab"
```

Invoke Design Compiler

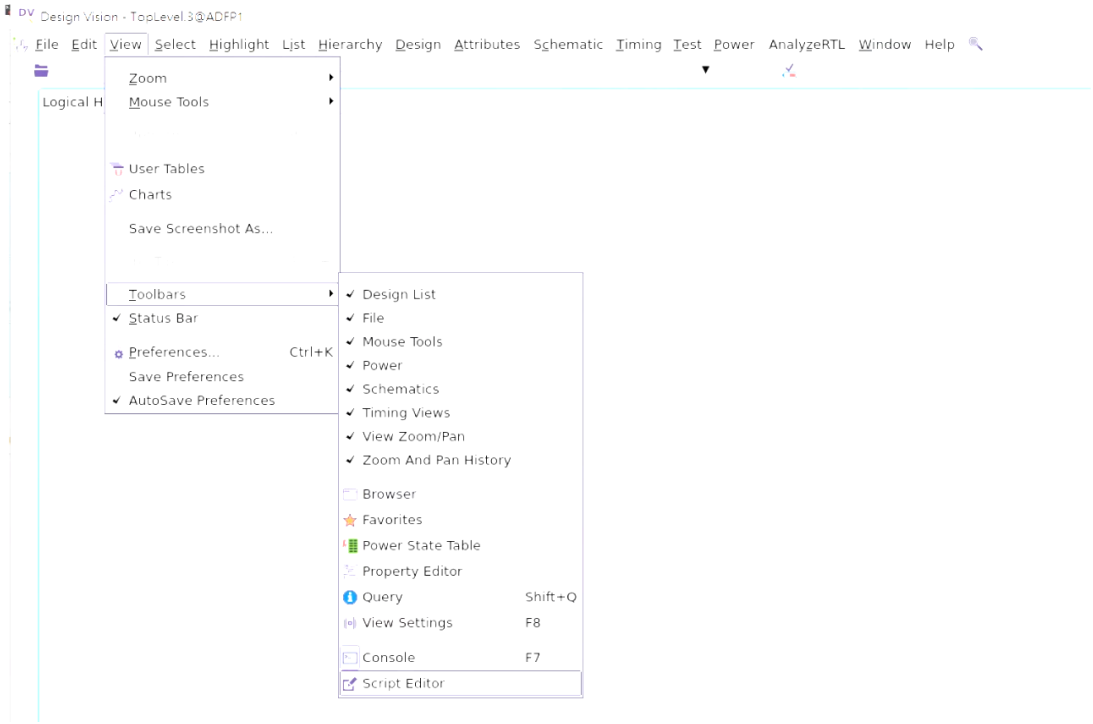
At the following path:

```
➢ cd ~/local_adfp_lab/Executable_Package/Lab/Work/sub_block/
  01_rtl_to_netlist/
```

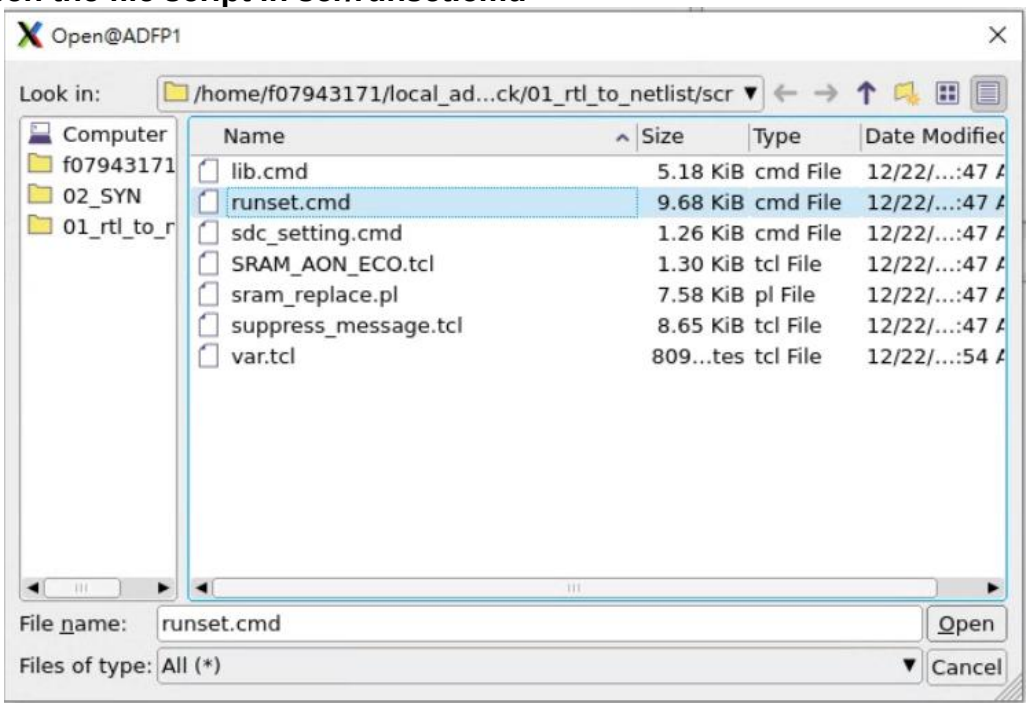
Invoke Design Compiler in topological and GUI mode.

```
➢ dc_shell-t -gui
```

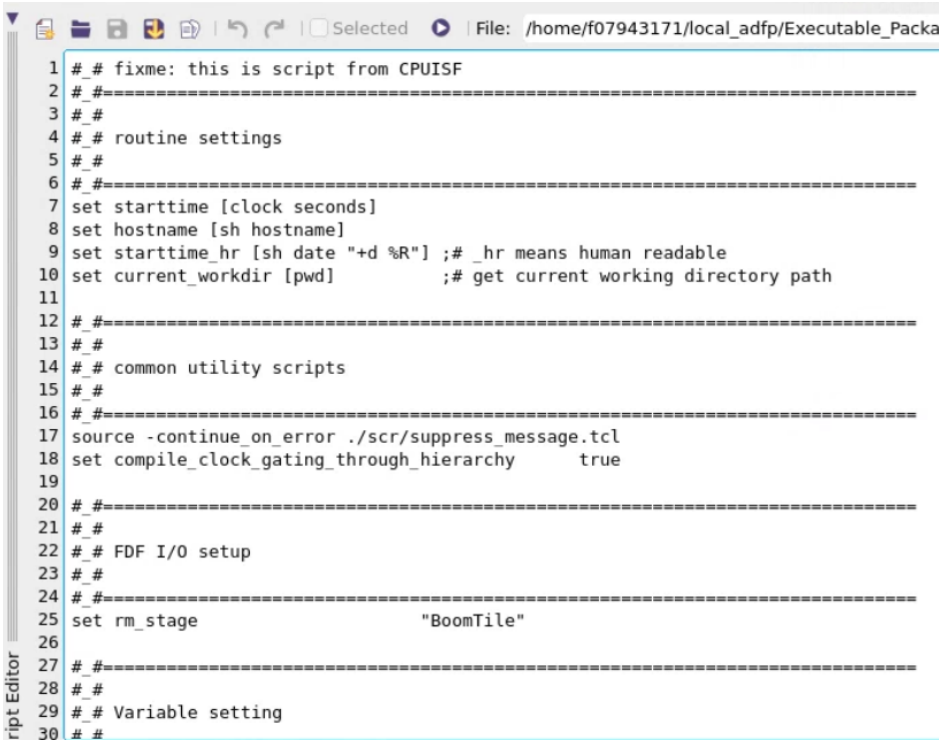
Click on View → Toolbars → Script Editor to invoke script editor. In this way, you can open a script and execute it line-by-line and by selection.



Open the file script in scr/runset.cmd



The opened script is then be viewed in the script editor.



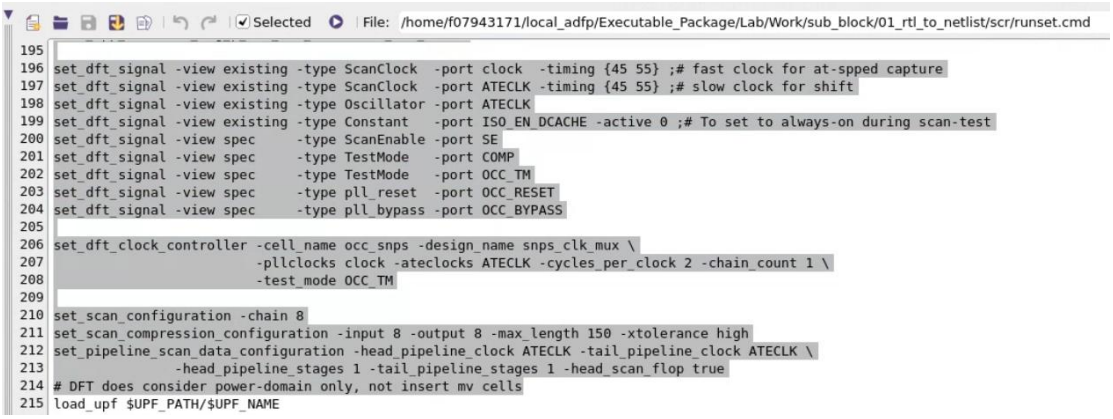

```
#_# fixme: this is script from CPUISF
#_#========================================================================
#_#
#_# routine settings
#_#
#_#========================================================================
set starttime [clock seconds]
set hostname [sh hostname]
set starttime_hr [sh date "+d %R"] ;# _hr means human readable
set current_workdir [pwd]          ;# get current working directory path

#_#========================================================================
#_#
#_# common utility scripts
#_#
#_#========================================================================
source -continue_on_error ./scr/suppress_message.tcl
set compile_clock_gating_through_hierarchy      true

#_#========================================================================
#_#
#_# FDF I/O setup
#_#
#_#========================================================================
set rm_stage                    "BoomTile"

#_#========================================================================
#_#
#_# Variable setting
#_#
```

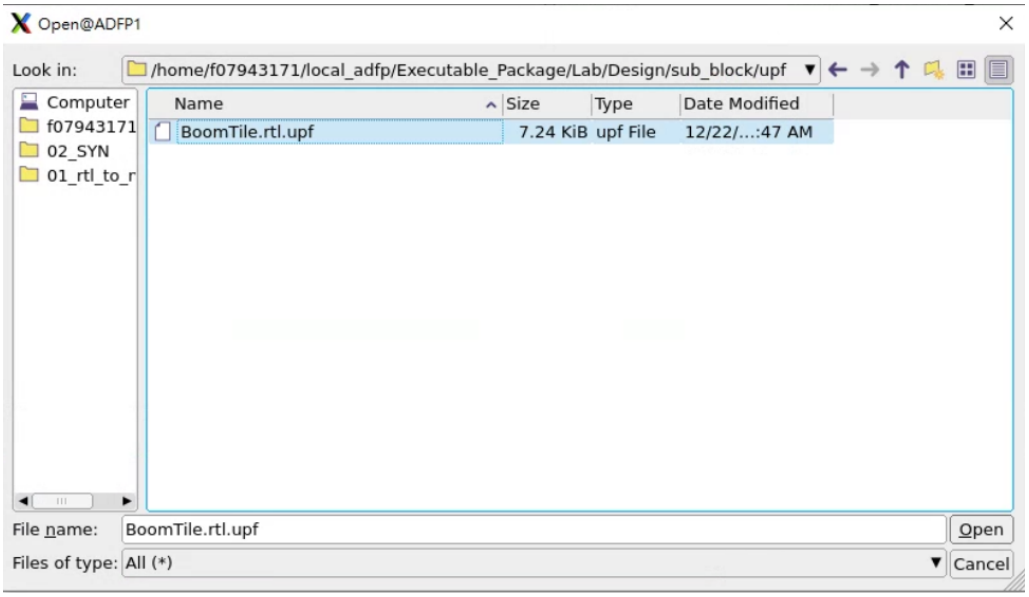
In this lab, we will only observe the UPF setting and the UPF diagram in the GUI mode. Select the script from the beginning to the line before load_upf command. Click on “Selected” and press the run button.

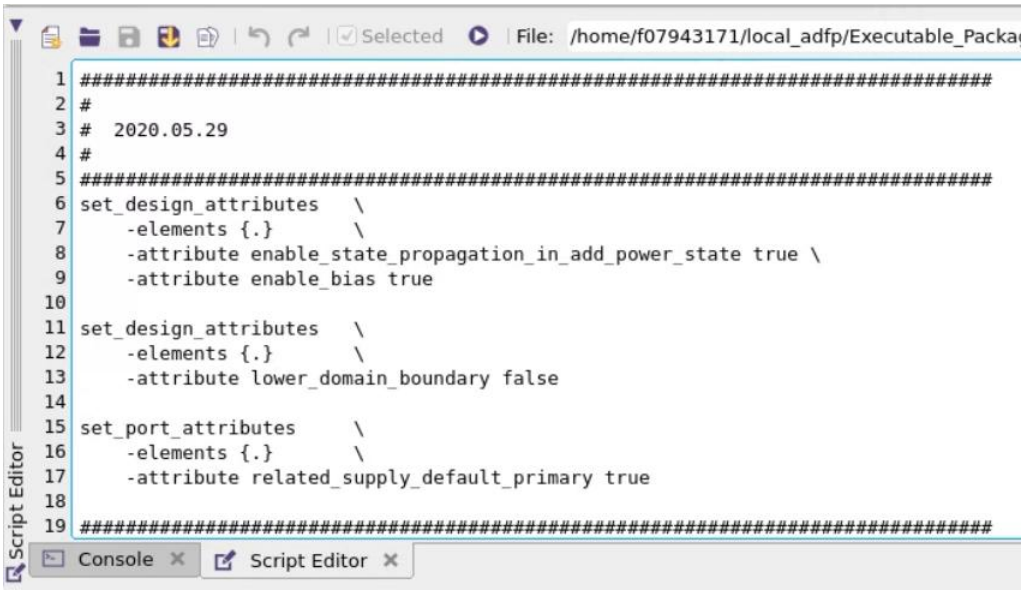



```
195
196 set_dft_signal -view existing -type ScanClock  -port clock  -timing {45 55} ;# fast clock for at-spped capture
197 set_dft_signal -view existing -type ScanClock  -port ATECLK -timing {45 55} ;# slow clock for shift
198 set_dft_signal -view existing -type Oscillator -port ATECLK
199 set_dft_signal -view existing -type Constant   -port ISO_EN_DCACHE -active 0 ;# To set to always-on during scan-test
200 set_dft_signal -view spec     -type ScanEnable -port SE
201 set_dft_signal -view spec     -type TestMode   -port COMP
202 set_dft_signal -view spec     -type TestMode   -port OCC_TM
203 set_dft_signal -view spec     -type pll_reset  -port OCC_RESET
204 set_dft_signal -view spec     -type pll_bypass -port OCC_BYPASS
205
206 set_dft_clock_controller -cell_name occ_snps -design_name snps_clk_mux \
207                          -pllclocks clock -ateclocks ATECLK -cycles_per_clock 2 -chain_count 1 \
208                          -test_mode OCC_TM
209
210 set_scan_configuration -chain 8
211 set_scan_compression_configuration -input 8 -output 8 -max_length 150 -xtolerance high
212 set_pipeline_scan_data_configuration -head_pipeline_clock ATECLK -tail_pipeline_clock ATECLK \
213                -head_pipeline_stages 1 -tail_pipeline_stages 1 -head_scan_flop true
214 # DFT does consider power-domain only, not insert mv cells
215 load_upf $UPF_PATH/$UPF_NAME
```

This step will cost several minutes (~10 min.) It first read the boom RISC-V RTL code, add ICG cells, create power gating control ports, compile it once and insert dft. (Not the key point of this lab, students can learn by themselves)

In script editor and open the upf file at the following path `~/local_adfp_lab/Executable_Package/Lab/Design/sub_block/upf/BoomtTile.rtl.upf`

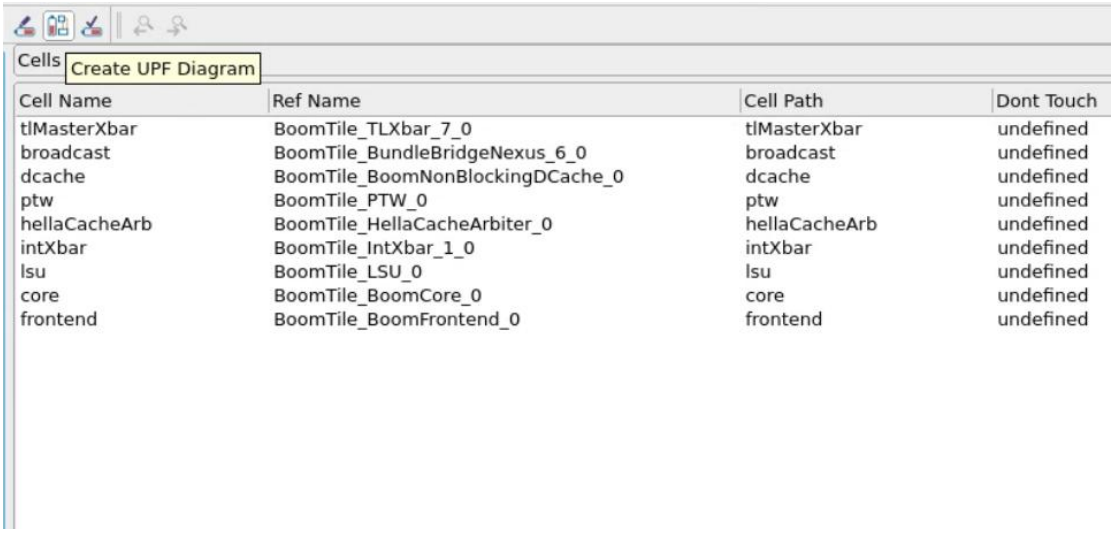




STEP 1: Create Power Domain

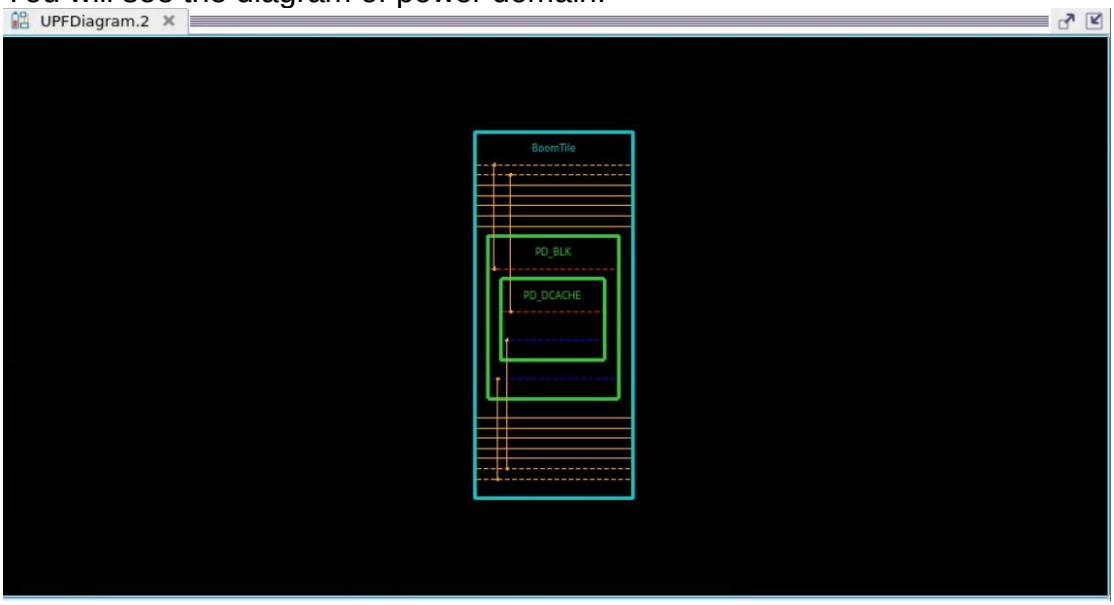
Select the code from the beginning to the end of create power domain and execute.

Open the toolbar → click on “Create UPF Diagram”



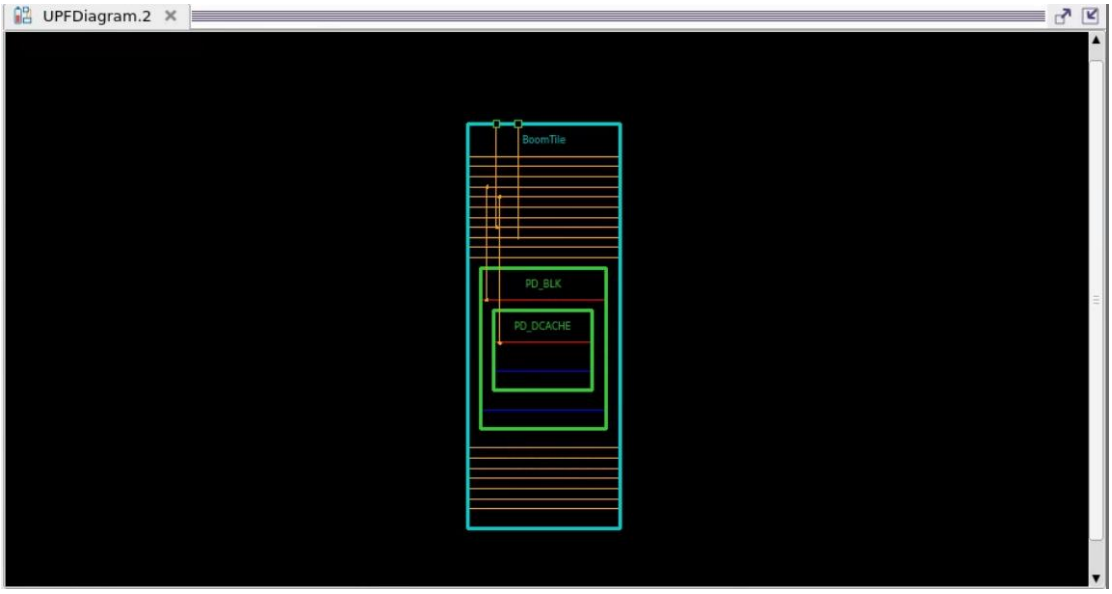
Cell Name	Ref Name	Cell Path	Dont Touch
tlMasterXbar	BoomTile_TLXbar_7_0	tlMasterXbar	undefined
broadcast	BoomTile_BundleBridgeNexus_6_0	broadcast	undefined
dcache	BoomTile_BoomNonBlockingDCache_0	dcache	undefined
ptw	BoomTile_PTW_0	ptw	undefined
hellaCacheArb	BoomTile_HellaCacheArbiter_0	hellaCacheArb	undefined
intXbar	BoomTile_IntXbar_1_0	intXbar	undefined
lsu	BoomTile_LSU_0	lsu	undefined
core	BoomTile_BoomCore_0	core	undefined
frontend	BoomTile_BoomFrontend_0	frontend	undefined

You will see the diagram of power domain.



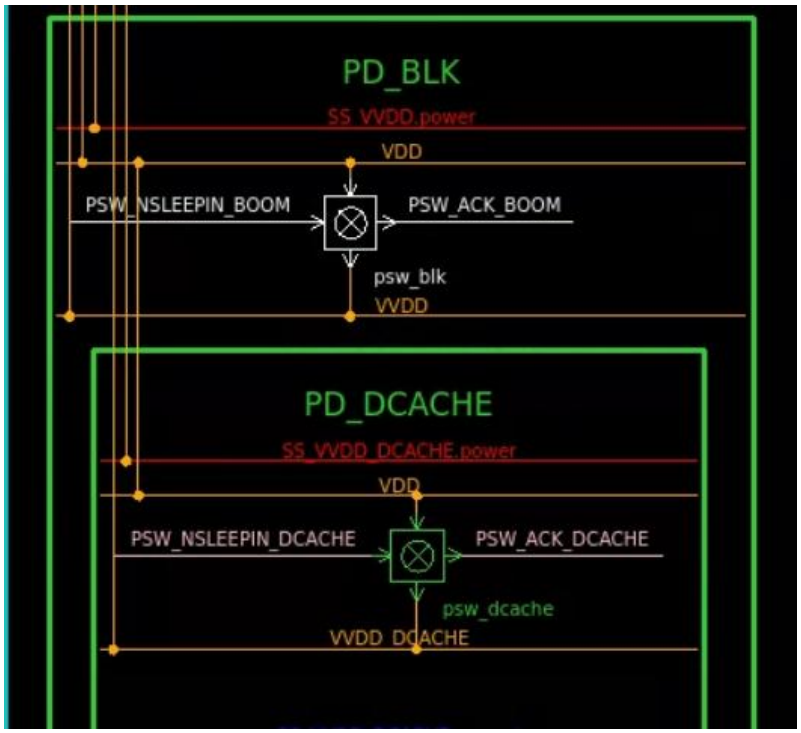
STEP 2: Assign Domain Supply and Connect Supply Port/Net

Select the code block “Assign Domain Supply” and “Power Supply Network” then execute them. Observe the updated upf diagram, and you will find the supply ports of VDD/VSS are created and the supply nets are connected to the PD_BLK and PD_DCACHE.



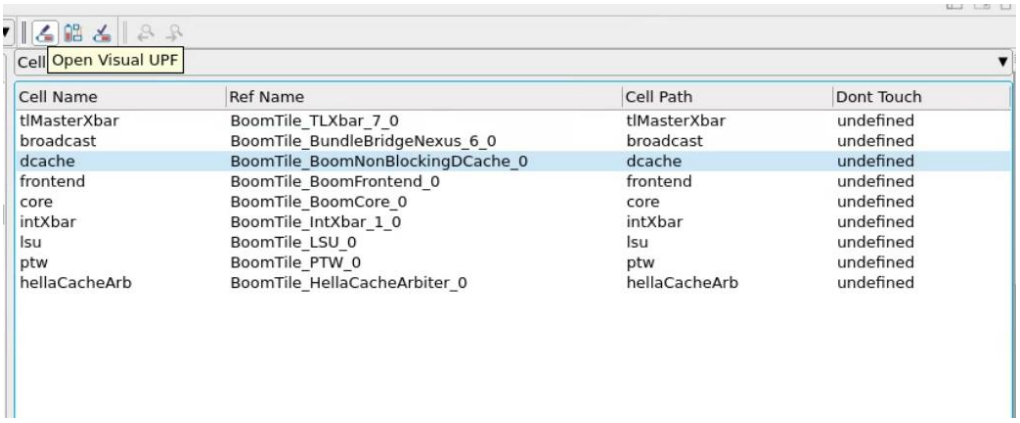
STEP 3: Add the Power Switches to the Two Domain

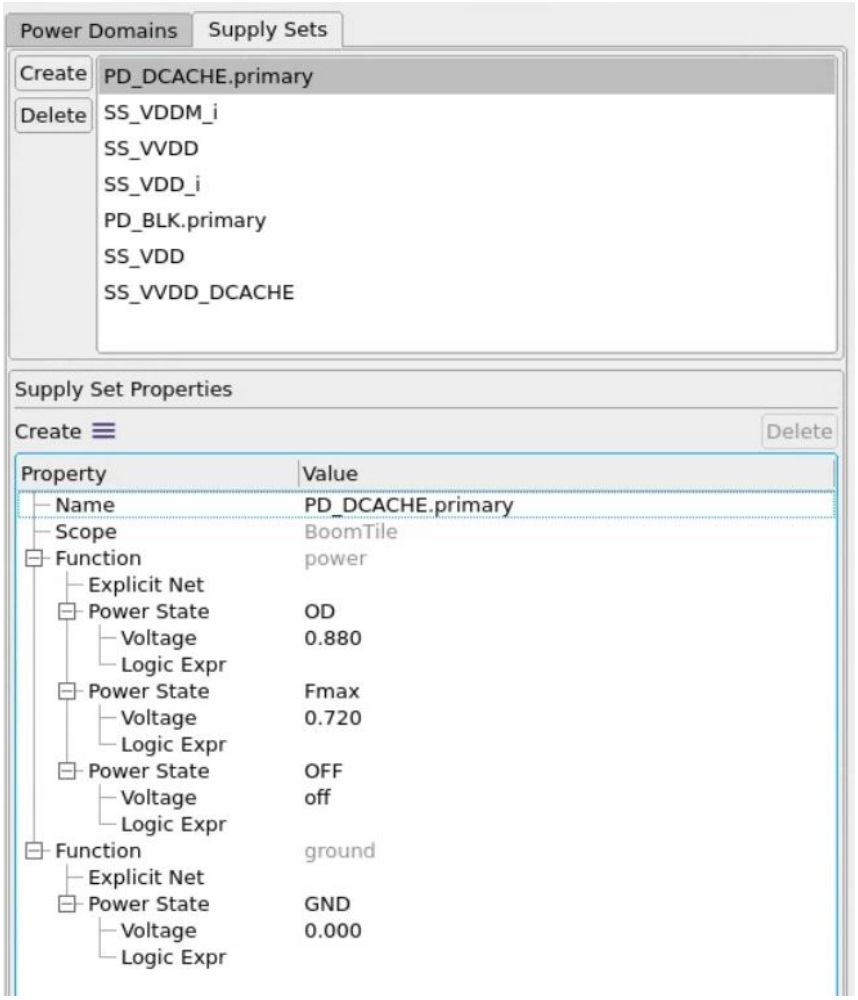
Select the code block “Power Gating”, and “Update Supply Set” then execute it. Observe the updated upf diagram, and you will find the power switches and its control signal. VVDD and VVDD_DCACHE are the virtual supply for VDD and VDD_CACHE.



STEP 4: Define Power State and Power State Table

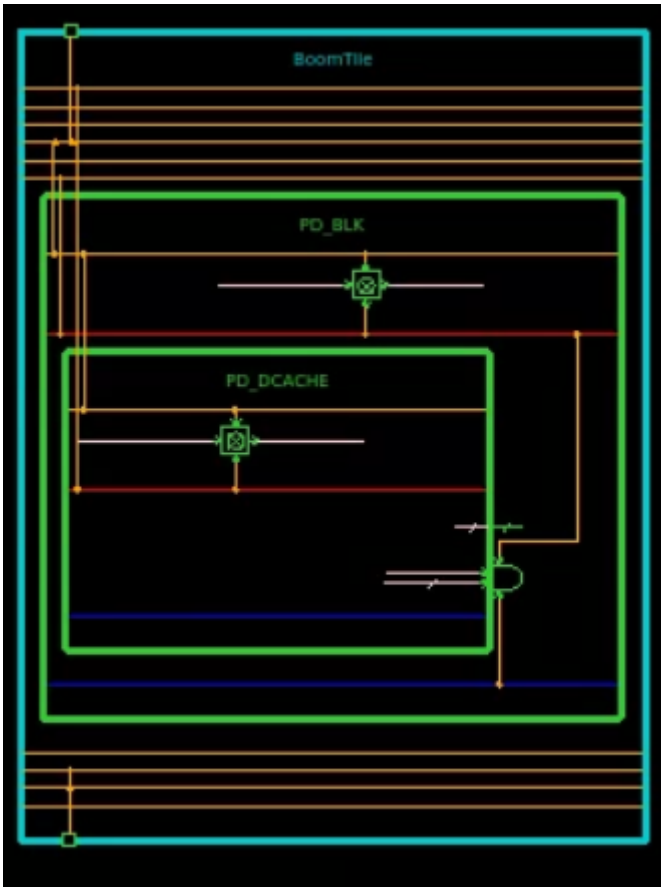
Select the code block “Define Power State”, and “Power State Table” then execute it. Check with the power state from supply set as follow:





STEP 5: Define Isolation Cell and Connect MACRO Supply Net

Select the code block “ISO”, and “Connect MACRO Supply net” then execute it. Check the updated upf diagram and you will see the isolation cells are added:



Checkpoints:

Please check with TAs before leaving this lab to make sure the following goals are accomplished and to get credits.

1. Show the snapshots of each step of the upf executions .
2. Show the power states of supply sets.

END of the LAB

Creator:
1st Edition: Yu-Chen Lo, 2024